23/3/26

Q. What are Utility classes in Java? Enlist any 5 methods of utility classes.

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Utility classes are special java classes that contain only static methods, static variables & private constructors. They are designed to provide reusable, helper functionality that doesn't require object instantiation.

- java.lang.Math class

(i) Math.abs() :- returns absolute (+ve) value of a number
eg:- Math.abs(-15); # 15

(ii) Math.pow() :- returns 1st no raised to 2nd no
eg:- Math.pow(2,3) # $2^3 = 8$.

(iii) Math.max() :- compares two nos & returns the larger one.
eg:- Math.max(10, 20) // 20

(iv) Math.min() :- compares two nos & returns the smaller one.
eg:- Math.min(10,20) // 10

(v) Math.sqrt() :- calculates square root of a number.
eg:- Math.sqrt(16); // 4

Q. What is a static method in java?
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The static keyword is used to create methods that belongs to the class rather than any specific instance of the class.

Features:-

- Static methods are associated with the class, not with individual objects. You can call them using the class name directly without creating an instance (object).

- Static methods can only directly access other static variables and static methods of the class.

- You can technically use an object, but it's not recommended (∵ static is used to save memory).

- Static methods are loaded into memory when the class is loaded, making them available immediately, even before object creation.

- Static methods are loaded once into memory at class loading time, rather than being created for each instance, making them memory efficient.

Q. Why main method is static in java?
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The main() method is static in java because the program needs to start running without creating an object first.

When you run a java program, the computer looks for the main() method directly on the class name itself, not on an object. If main() were not static, the computer would have to create an object before it could start the program, which would be unnecessary & complicated.

By making it static, the program can begin immediately as soon as the class is loaded, without needing to do anything extra.

Q. Can we call instance method, or things from static method directly?

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No, you cannot call instance (non-static) things directly from a static method in java. A static method belongs to the class itself, not to a specific object (instance) of the class, & it exists even before any objects are created.
To access instance things from a static context, you must have a reference to an object of the class.

Q. Can we call static method to instance method directly?

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Yes, we can call static methods / things to an instance method directly by classname & there is no need to create objects to access static things.
eg. Demo.m1(); ClassName.methodName();

Q. Any built-in class where we can find static / normal blocks?

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(i) java.util.System

(ii) java.lang.Boolean

Q. Any built-in class where we can find static final variables?

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java.lang.Boolean